Open Hashing

n = Number in
m = Total

$i_k = hash(id_j)$
$i_j = hash(id_j)$

$i_j = i_k$ → Collision

$i_k = (i_k + r \times p) \bmod m + 1$

$n < m \qquad \alpha = \frac{n}{m}$

Why should m be prime?

Chained Hashing

linked lists are fast due to pointer.

$\alpha = \frac{n}{m} \Rightarrow$ if $\alpha = \frac{2n}{m} = 2\alpha$

A coeff > 1 is possible.

NB: α describes mean length of the chain.

a[m+1] has m positions and n entries.

- $1 + \frac{\alpha+1}{2}$ (1 for first probe/link)
- $O(\alpha)$ (Average length is proportional to alpha)

($O(\alpha)$ is mathematically a set {})

So we can say an algorithm is

$\in O(\alpha)$

Hans Peter Luhn → Invented Hashing at IBM

How does RSA calculate your ID (Luhn Hashing)